DATABASE-SYSTEMS

Lecture - 1

By

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What we are going to learn in this subject?

The basic concepts of databases, why we need them, how it can be subdivided, why it is important, what we can achieve by this course?

What is Database

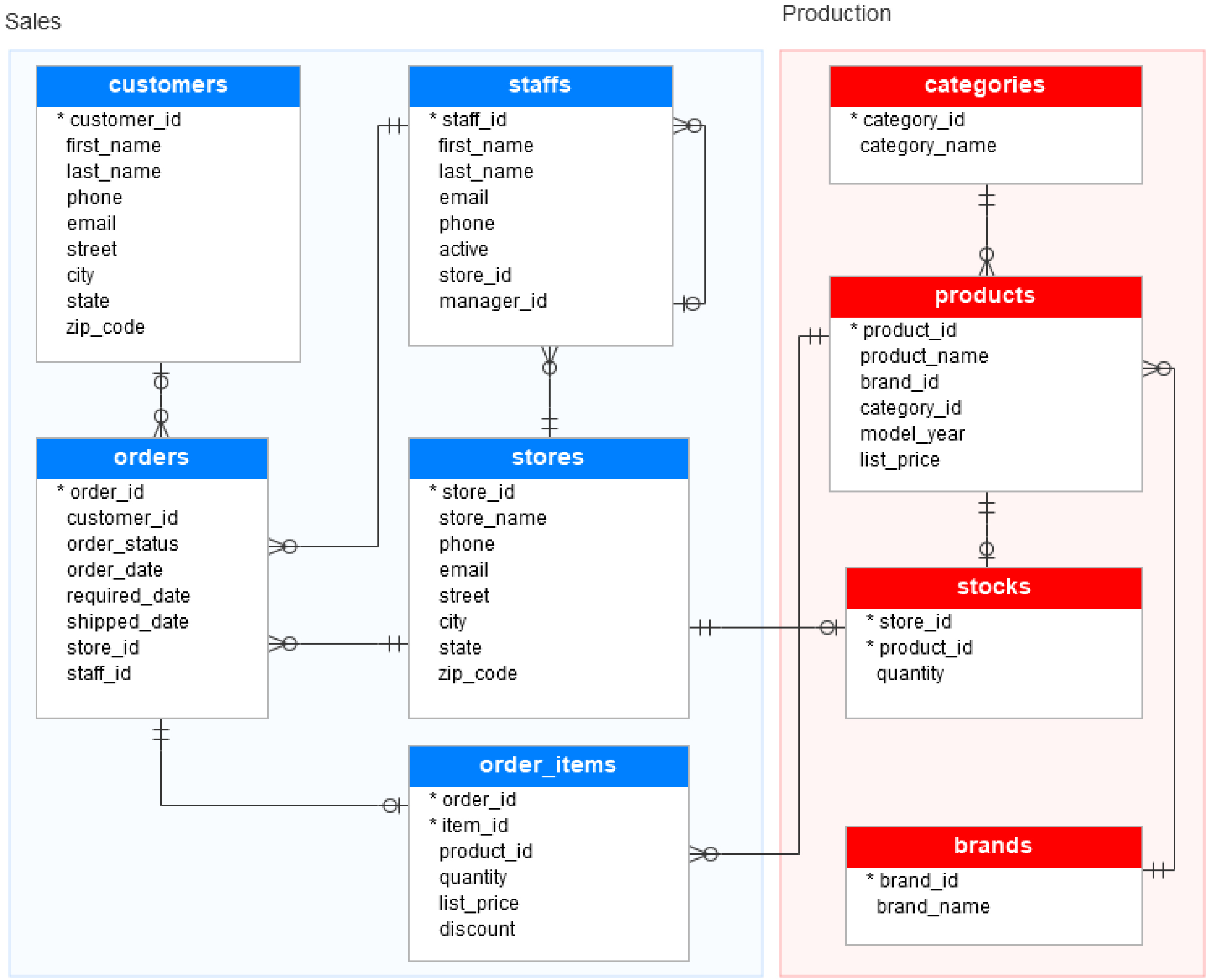
What comes in our mind?

Air - Base

Data - Base

What is database

But actually it is?



Transparent table: YSTRAVELAG

Short description: Customer table: Travel agency

Fields	Key	Init.	Field type	Date	Length	DecPl.	Check table	Short description
MANDT	✓	✓	S_MANDT	CLNT	3	0	T000	Client for WB training data model
AGENYNUM	✓	✓	S_AGNCYNUM	NUMC	8	0	SBUSPART	Number of travel agency
NAME		✓	S_AGNCYNAM	CHAR	25	0		Name of travel agency
STREET			S_STREET	CHAR	30	0		Street for WB training data model
POSTBOX			S_POSTBOX	CHAR	10	0		PO box for WB training data mod
POSTCODE		✓	POSTCODE	CHAR	10	0		Zip code for WB training data mo
CITY		✓	CITY	CHAR	25	0		City for WB training data model
COUNTRY			S_COUNTRY	CHAR	3	0		Country code for WB training data
REGION			S_REGION	CHAR	3	0		Region of a country
TELEPHONE		✓	S_PHONED	CHAR	30	0		Phone number of a customer
URL			S_URL	CHAR	50	0		URL of travel agency's home page
LANGU			SPRAS	LANG	1	0	T002	Language key

Data Vs Information & its types

- **Data:** A collection of raw facts and figures is known as data.
- **Types of Data**
  - **Numeric Data**
  - **Alphabetic Data**
  - **Alphanumeric Data**
  - **Image Data**
  - **Audio Data**
  - **Video Data**

Information: The processed data is known as information. A useful or meaningful data is also known as information.

What is “database” & “database systems”

“A ***DATABASE*** *is an organized collection of structured information, or data, typically stored electronically in a computer system.”*

“A database is usually controlled by a ***DATABASE MANAGEMENT SYSTEM (DBMS)****. Together, the data and the DBMS, along with the applications that are associated with them, are referred to as a database system, often shortened to just database.”*

What is “database” & “database systems”

- Database involves many thing but some of them following:
  - Store structure of Data
  - Types of data
  - Modeling of Data
  - Management of Data
  - etc

Difference b/w database and data structure

V/S

Difference b/w database and data structure

Database	Data Structure
It is an **organized collection** of data.	It is a special format for storing data to serve a particular purpose.
It **is used to access** the data and manage it easily.	It is used for efficiency and to reduce the complexities of the program.
Structured query language (SQL) is used to perform operations on the data in a database. Database Management System **(DBMS)** is used to manage the database.	Programming languages like C, C++, Java, Python are used to perform operations using data Structures.
It is a **Non-volatile** memory.	It is a volatile memory.
Types: Relational database, NOSQL database, Centralized database, Hierarchical database are some types of databases.	**Types:** Linear data structure and non-linear data structure are the types of data structures.
Example: MySQL, SQL Server, Oracle, MongoDB, SQLite and etc.	**Example:** Array, Linked List, Stack, Queue, Tree, Graph.

Difference b/w database system and file system

File System	Database Management System
File system is a software that **manages** and organizes the **files** in a storage medium within a computer.	DBMS is a software for **managing the database**.
Redundant data can be present in a file system.	In DBMS there is **no redundant data**.
It **doesn’t provide backup** and recovery of data if it is lost.	It **provides backup and recovery** of data even if it is lost.
There is **no efficient query processing** in file system.	**Efficient query processing** is there in DBMS.
There is **less data consistency** in file system.	There is **more data consistency** because of the process of normalization.
It is **less complex** as compared to DBMS.	It has **more complexity in handling** as compared to file system.
File systems provide **less security in comparison** to DBMS.	DBMS has **more security mechanisms** as compared to file system.
It is **less expensive** than DBMS.	It has a **comparatively higher cost** than a file system.

Concepts of Tables in Database

Table is the fundamental object of the database structure. The basic purpose of a table is to store data. A table consists of rows and columns.

Serial No.	Name	Qualification	Email
1	Farooq	Mphil	farooq11@hotmail.com
2	Ejaz	PhD	Ejaz_1128@yahoo.com
3	Usman	BS	dbprogrammer22@gmail.com

Row/Record/tuple: Rows are the horizontal part of the table. A single row is known as record or tuple. Table in previous slide has 3 records.

Column/field/attribute: Columns are the vertical part of the table. A single column is known as field or attribute. Above table has 4 fields or attributes.

Relation: In relation model, data is stored in the form of relations. Relation is another name used for table.

Concepts of Users in Database

End users (EU)

Use the database system to achieve some goal

Application developers (AD)

Write software to allow end users to interface with the database system

Database Administrator (DBA)

Designs & manages the database system/Provide technical support

Database systems programmer

Writes the database software itself

Database Management System

DBMS is a system software for creating and managing databases.

A software that interacts with end users, applications, and the database itself to store and analyze the data

DBMS provides:

efficient, reliable, convenient and safe (don't loss or override or corrupt) multi-user (concurrency control) storage of and access to massive amount of persistent data.

Database Management System

- A set of programs which manages and control database is known as Database Management System.
- List of functions of DBMS:
  - Create database
  - Create tables
  - Create structures
  - Read database data
  - Update database data
  - Maintain database structures
  - Enforce rules
  - Perform Security
  - Perform Backup and Recovery

Examples of Database Management System

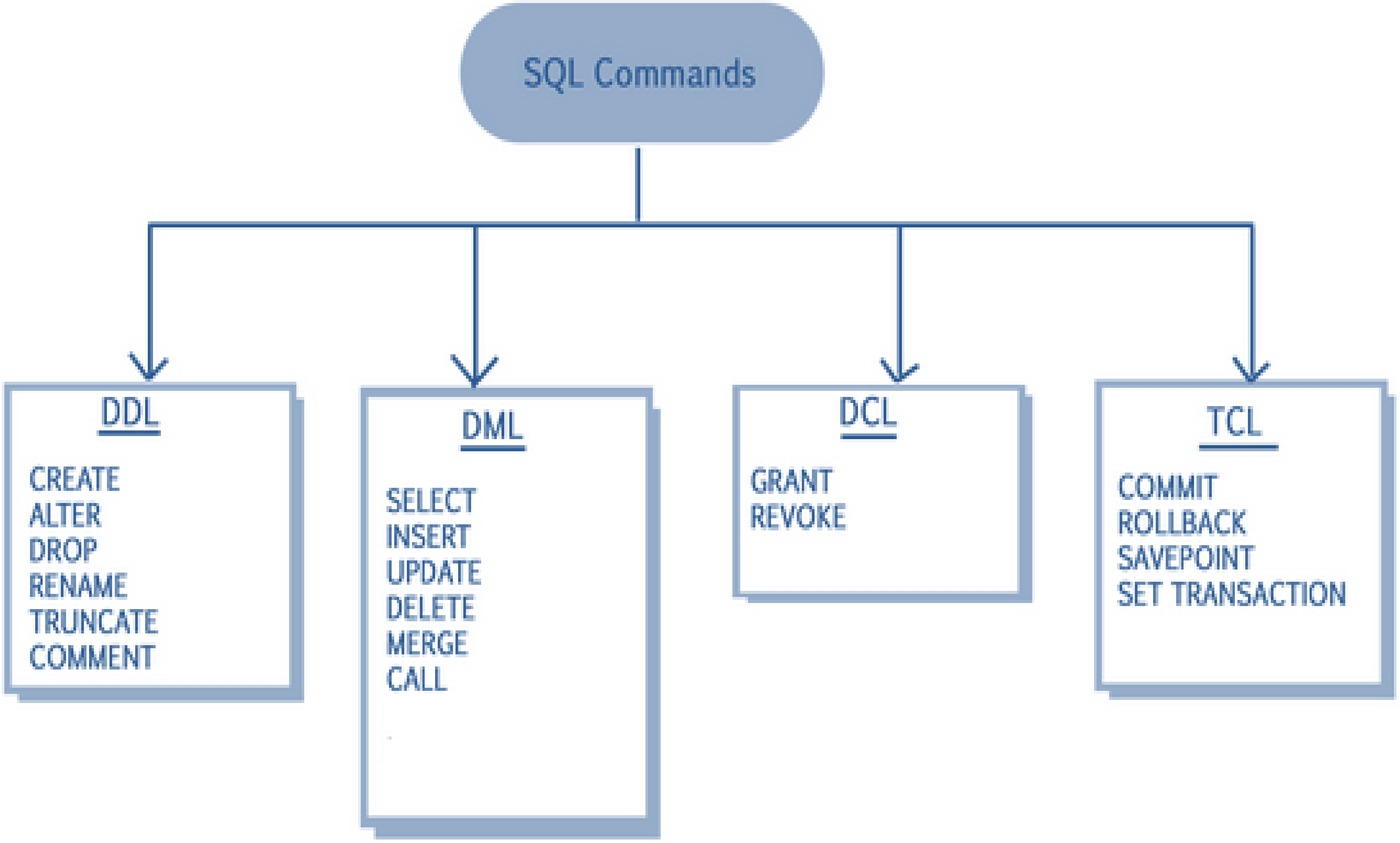
- Web indexes
- Library catalogues
- Medical records
- Bank accounts
- Stock control
- Personnel systems
- Product catalogues
- Telephone directories

- Train timetables
- Airline bookings
- Credit card details
- Student records
- Customer histories
- Stock market prices
- Discussion boards
- and so on…

What Database Management System Do

- Provides users with
  - Data definition language (DDL)
  - Data manipulation language (DML)
  - Data control language (DCL)
  - Transaction Control Language (TCL)
- Often these are all the same language

What Database Management System Do



Data Definition Language

- These SQL commands are mainly categorized into four categories as discussed below:
- **DDL(Data Definition Language) :** DDL or Data Definition Language actually consists of the SQL commands that can be used to define the database schema. It simply deals with descriptions of the database schema and is used to create and modify the structure of database objects in database. **Examples of DDL commands:**
  - **CREATE** – is used to create the database or its objects
  - **DROP** – is used to delete objects from the database.
  - **ALTER**-is used to alter the structure of the database.
  - **TRUNCATE**–is used to remove all records from a table, including all spaces allocated for the records are removed.
  - **COMMENT** –is used to add comments to the data dictionary.
  - **RENAME** –is used to rename an object existing in the database.

Data Manipulation Language

- DML(Data Manipulation Language) : The SQL commands that deals with the manipulation of data present in database belong to DML or Data Manipulation Language and this includes most of the SQL statements. Examples of DML:
- SELECT – is used to retrieve data from the a database.
- INSERT – is used to insert data into a table.
- UPDATE – is used to update existing data within a table.
- DELETE – is used to delete records from a database table.

DCL and TCL

- **DCL(Data Control Language) :** DCL includes commands such as GRANT and REVOKE which mainly deals with the rights, permissions and other controls of the database system. **Examples of DCL commands:**
  - **GRANT**-gives user's access privileges to database.
  - **REVOKE**-withdraw user's access privileges given by using the GRANT command.
- **TCL(Transaction Control Language) :** TCL commands deals with the transaction within the database. **Examples of TCL commands:**
  - **COMMIT**– commits a Transaction.
  - **ROLLBACK**– rollbacks a transaction in case of any error occurs.

What DBMS Does

- DBMS provides
  - Persistence
  - Concurrency (Multi tasking)
  - Integrity
  - Security
  - Data independence
- Data Dictionary
  - Describes the database itself

Data Dictionary - Metadata

- The dictionary or catalogue stores information about the database itself
- This is data about data or ‘metadata’
- Almost every aspect of the DBMS uses the dictionary
- The dictionary holds
    - Descriptions of database objects (tables, users, rules, views, indexes,…)
    - Information about who is using which data (locks)
    - Schemas and mappings

Database Environment

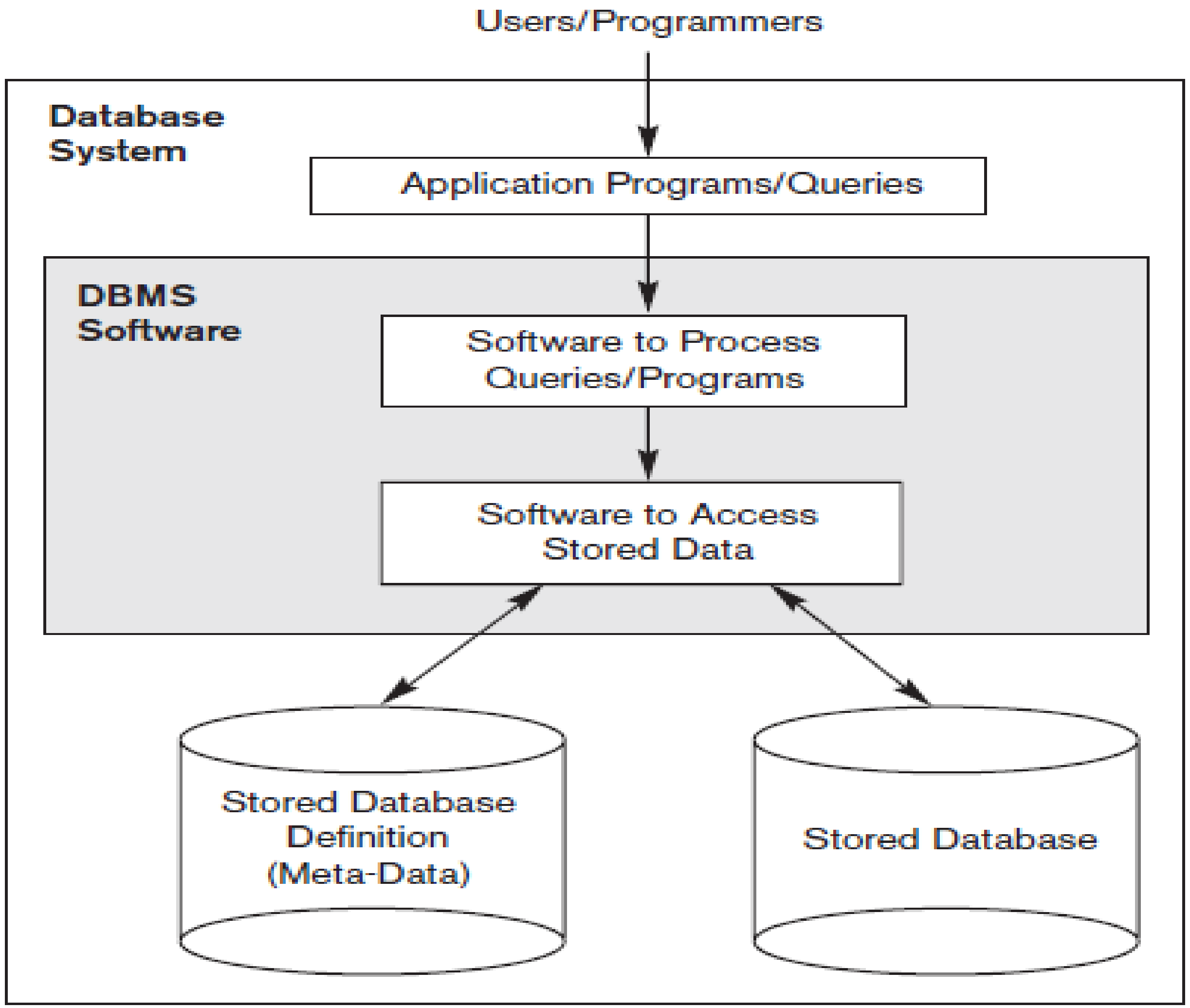


Figure 1.1
A simplified database system environment.

Database Environment

Backend / Developer Side

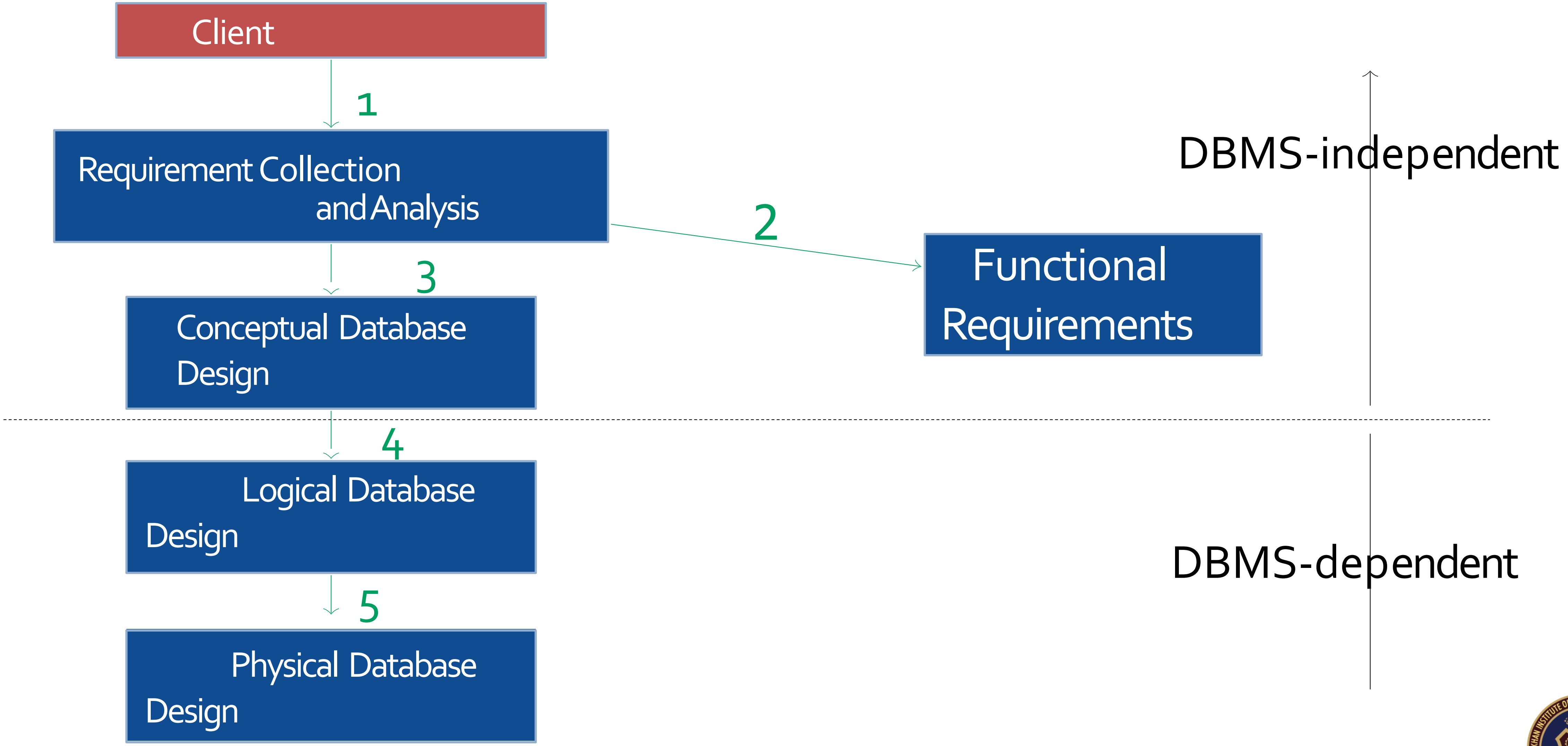
ID	FirstName	LastName	UserName	EmailID	PasswordHash
1	Justine	Henderson	justine_henderson	justine_henderson@mail-server.com	BLOB
2	Abdullah	Lang	abdullah_lang	abdullah_lang@mail-server.com	BLOB
3	Marcus	Cruz	marcus_cruz	marcus_cruz@mail-server.com	BLOB
4	Thalia	Cobb	thalia_cobb	thalia_cobb@mail-server.com	BLOB
5	Mathias	Little	mathias_little	mathias_little@mail-server.com	BLOB
6	Eddie	Randolph	eddie_randolph	eddie_randolph@mail-server.com	BLOB

Front End / User Side

Phases of Database Design

- Requirement Collection and Analysis
- Functional Requirements
- Conceptual Database Design
- Logical Database Design
- Physical Database Design

Phases of Database Design



Requirement Collection and Analysis

Database designers interview clients to understand and document their requirements

The business is a sweet factory called 'Sweets R Us'. The business buys raw ingredients from several suppliers and keeps a monthly record of the purchases. Sweets R Us has several employees who prepare the ingredients or make sweets in three of the company's departments. The products are sold to a number of retail stores, and the sales are recorded in an inventory on a monthly basis.

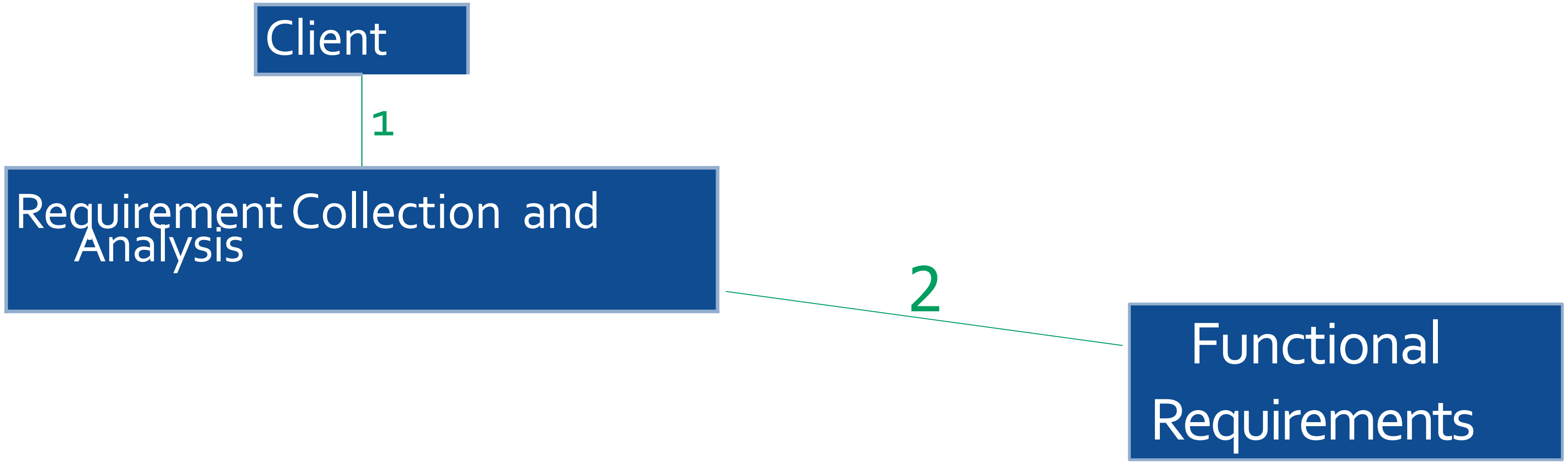
The company would like to keep track of purchases of raw ingredients and their suppliers, as well as employees and their wages. There are 20 employees in the company, and 3 departments. The employees specialize in different area of the sweet manufacturing process and have different job descriptions. The company sells their products to a number of different retailers in various cities. They would like to keep track of this information.

As far as keeping track of sales and purchases is concerned, the company would like to keep track of the value and date of sales and purchases as well who bought or sold what to them.

The company is loosing a lot of money and they would like to be able to see where costs could be cut. One example would be to examine how much they could save by reducing the highest paid employees wages, with the idea that they could let those employees go, and replace them with lower skilled employees.

They would also like to keep track of where employees live with the aim of letting go of employees who live too far away. Another request is to be able to locate retailers from various cities so they can reduce costs by having one single delivery system to each city, rather than have multiple deliveries there.

Functional Requirements



What kind of queries will be applied to the database? transactional queries or analytical queries?

Conceptual Schema

Client

1

Requirement Collection and Analysis

3

Conceptual Database Design

Models all of the information gathered in phase1

Includes detailed descriptions of entities and relationships

Conceptual Schema={entities, attributes, relationships}

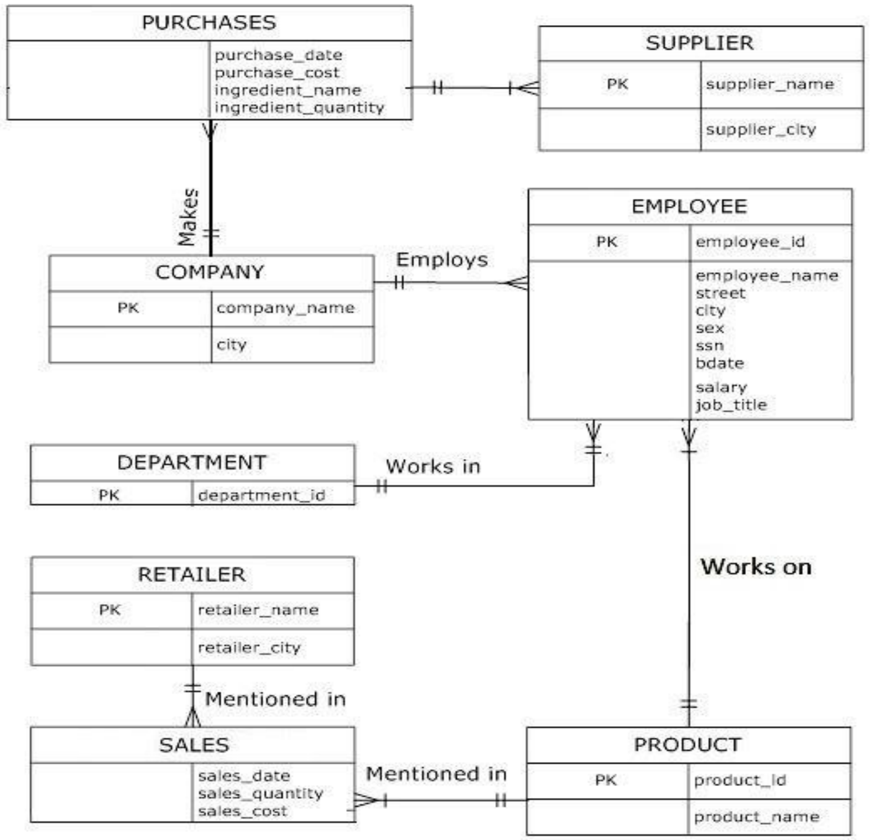


Figure: Conceptual Schema

Logical Database Schema

Client

↓ 1

Requirement Collection and Analysis

↓ 3

Conceptual Database Design

↓ 4

Logical Database Design

Constraint

Enforces limit to the data that can be inserted/updated/deleted from a table (e.g. attribute X cannot have duplicate values; attribute Y cannot have null values)

Logical Schema = {entities, attributes, constraints, data types, relationships}

Describes what data will be stored (e.g. what tables are used, what data types are used, what constraints are applied)

Logical Schema

COMPANY (company name, city);
PURCHASES (company name, supplier name, purchase_date, purchase_cost, ingredient_name, ingredient_quantity)
SUPPLIER (supplier name, supplier_city)
EMPLOYEE (employee id, employee_name, street, city, sex, ssn, bdate)
WORKS (employee id, department id, product id, job_title, salary)
DEPARTMENT (department id, company_name)
PRODUCT (product id, product_name)
SALES (product id, retailer name, sales_date, sales_quantity, sales_cost)
RETAILER (retailer name, retailer_city)

	A	B	C	D	E	F	G	H	I
1	TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	FORMAT	RANGE	REQUIRED	PK/FK	REFERENCE
2	COMPANY	company_name	Company Name	varchar2 (19)	Xxxxxxx		Y	PK	
		city	City where the company is based	varchar2 (19)	Xxxxxxx				
3	PURCHASES	company_name	Company Name	varchar2 (19)	Xxxxxxx			PK FK	COMPANY
		supplier_name	Supplier Company Name	varchar2 (19)	Xxxxxxx	-		PK FK	SUPPLIER
		purchase_date	Date when the purchase was made	date	dd-mm-yy	-			
		purchase_cost	How much the order cost	number (6,2)	9999	1.00 - 9999.99			
		ingredient_name	What was bought	varchar2 (19)	Xxxxxxx	-			
		ingredient_quantit	How much was bought	number (3,0)	999	1 - 999			
4	SUPPLIER	supplier_name	Supplier Company Name	varchar2 (19)	Xxxxxxx		Y	PK	
		supplier_city	City where the supplier is based	varchar2 (19)	Xxxxxxx				
5	EMPLOYEE	employee_id	Employee id number	char (4) not null	999999999	1111 - 9999	Y	PK	
		employee_name	Full name of employee	varchar2 (19)	Xxxxxxx	-			
		street	Employees street address	varchar2 (19)	Xxxxxxx	-			
		city	City where employee lives	varchar2 (19)	Xxxxxxx	-			
		sex	Gender of employee	char (3)	X	-			
		ssn	Social security number	char (5)	999999999	11111 - 99999			
		bdate	Date of birth	date	dd-mm-yy				
6	WORKS	employee_id	Employee id number	char (4) not null	999999999	1111 - 9999	Y	PK FK	EMPLOYEE
		job_title	Title of job	varchar2 (19)	Xxxxxxx	-		-	-
		department_id	Id of department where he works	char (3)	999	111 - 999		PK FK	DEPARTMEN
		product_id	Id number of product he makes	char (5)	99999	11111 - 99999		PK FK	PRODUCT
		salary	Annual salary	number (8,2)	999999.99	0.00 - 999, 999.99			
7	DEPARTMENT	department_id	Id of department	char (3)	999	111 - 999	Y	PK	
		company_name	Name of company	varchar2 (19)	Xxxxxxx				
8	PRODUCT	product_id	Final Product Id	char (5)	99999		Y	PK	
		product_name	Name of final product	varchar2 (19)	Xxxxxxx	11111 - 99999			
9	SALES	product_id	Final Product Id	char (5)	99999	11111 - 99999		PK FK	PRODUCT
		retailer_name	Name of company that bought it	varchar2 (19)	Xxxxxxx	-		PK FK	RETAILER
		sales_date	Date when product was sold	date	dd-mm-yy	-			
		sales_quantity	How much was sold to retailer	number (4,0)	9999	1 - 9999			
		sales_cost	Total amount charged	number (6,2)	9999.99	0.00 - 9999.99			
10	RETAILER	retailer_name	Retailer name	varchar2 (19)	Xxxxxxx		Y	PK	
		retailer_city	Retailer's city	varchar2 (19)	Xxxxxxx				

Physical Database Schema

Client

1

Requirement Collection and Analysis

2

Functional Requirements

3

Conceptual Database Design

4

Logical Database Design

5

Physical Database Design

SQL code used to build database describes how data will be stored (i.e. define index structures and storage layout)

Physical Database Schema

```
create table COMPANY (
company_name varchar2(19) not null,
city varchar2(19),
CONSTRAINT COMPANY_company_name_pk
PRIMARY KEY(company_name)
);
.
.
.
.
create table SALES (
product_id  char (5) not null,
retailer_name varchar2 (19) not null,
sales_date date,
sales_quantity number (4,0),
sales_cost number (6,2),
CONSTRAINT SALES_product_id_fk FOREIGN KEY(product_id)
REFERENCES PRODUCT(product_id),
CONSTRAINT SALES_retailer_name_fk FOREIGN KEY(retailer_name)
REFERENCES RETAILER(retailer_name)
CONSTRAINT SALES_product_id_pk
PRIMARY KEY(product_id, retailer_name)
);
```

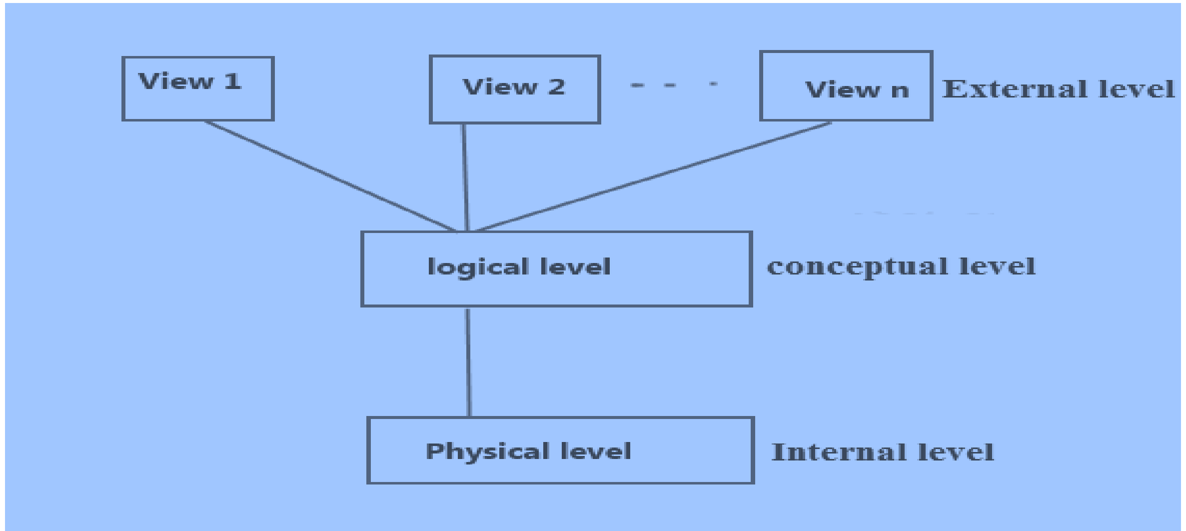
Figure: Internal Schema

Internal Schema = {entities, attributes, relationships, data types, constraints, indexes}

Levels of Abstraction

To ease the user interaction with database, the developers hide internal irrelevant details from users. This process of hiding irrelevant details from user is called data abstraction.

- There three types of level
  - External level/View level
  - Logical level/Conceptual level
  - Physical level/Internal level



Three Levels of data abstraction

View Level/ External Level

Highest level of data abstraction. This describes the user interaction with database system.

At **external/view level**, user just interact with system with the help of GUI and enter the details at the screen, they are not aware of how the data is stored and what data is stored; such details are hidden from them.

Logical / Conceptual Level

Logical level: This is the middle level of 3-level data abstraction architecture. It describes what data is stored in database.

It describes what will be the entities, what will be their attributes and relationships between them.

Physical / Internal Level

This is the lowest level of data abstraction. It describes how data is actually stored in database and what type of data will be stored.

Let's say we are storing customer information in a customer table. At **physical level** these records can be described as blocks of storage (bytes, gigabytes, terabytes etc.) in memory.

Installation

- Postgres
- Mongo-DB / Firebase
- Python
- Pycharm